

# Vaja 7: Nejc Dolinar

**Cilj vaje:** Vzpostavitev varnega okolja s spletnim strežnikom v javnem podomrežju in podatkovno bazo v privatnem podomrežju ter njuna medsebojna povezava.

## 1. Konfiguracija EC2 instanc

V tem koraku smo pripravili dve instanci:

- **Vaja7-Public (Spletni strežnik):** Nahaja se v javnem podomrežju z dodeljenim javnim IPv4 naslovom (3.71.8.123).
- **vaja7-db-primary (Podatkovni strežnik):** Nahaja se v privatnem podomrežju brez javnega dostopa, dostopna le preko lokalnega IP naslova (192.168.0.140).

Instances (1/3) info

Save filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

| Name               | Instance ID         | Instance state | Instance type | Status check      | Availability Zone | Public IPv4 DNS           | Public IPv4 ... | Elastic IP | IPv6 IPs | Monitoring |
|--------------------|---------------------|----------------|---------------|-------------------|-------------------|---------------------------|-----------------|------------|----------|------------|
| vaja7-db-secondary | i-077b44f66436ad491 | Running        | t3.micro      | 3/3 checks passed | eu-central-1b     | -                         | -               | -          | -        | disabled   |
| Vaja7-Public       | i-066d0deb1f875ee2f | Running        | t3.micro      | 3/3 checks passed | eu-central-1a     | ec2-3-71-8-123.eu-cent... | 3.71.8.123      | -          | -        | disabled   |
| vaja7-db-primary   | i-0f2256fa2aabfe2dd | Running        | t3.micro      | 3/3 checks passed | eu-central-1a     | -                         | -               | -          | -        | disabled   |

i-066d0deb1f875ee2f (Vaja7-Public)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary info

Instance ID

i-066d0deb1f875ee2f

IPv6 address

-

Hostname type

IP name: ip-192-168-0-10.eu-central-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

3.71.8.123 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

▼ Instance details info

AMI ID

Monitoring

Platform details

Public IPv4 address

3.71.8.123 [open address]

Instance state

Running

Private IP DNS name (IPv4 only)

ip-192-168-0-10.eu-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0735033c90d61e0f7 (vaja7-vpc)

Subnet ID

subnet-0d12d6206b893344c (vaja7-subnet-public1-eu-central-1a)

Instance ARN

arn:aws:ec2:eu-central-1:159858924832:instance/i-066d0deb1f875ee2f

Private IPv4 addresses

192.168.0.10

Public DNS

ec2-3-71-8-123.eu-central-1.compute.amazonaws.com [open address]

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

i-0f2256fa2aabfe2dd (vaja7-db-primary)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary info

Instance ID

i-0f2256fa2aabfe2dd

IPv6 address

-

Hostname type

IP name: ip-192-168-0-140.eu-central-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

-

IAM role

-

IMDSv2

Required

Operator

-

▼ Instance details info

AMI ID

Monitoring

Platform details

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-192-168-0-140.eu-central-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0735033c90d61e0f7 (vaja7-vpc)

Subnet ID

subnet-0927e7b20b3739cb (vaja7-subnet-private1-eu-central-1a)

Instance ARN

arn:aws:ec2:eu-central-1:159858924832:instance/i-0f2256fa2aabfe2dd

Private IPv4 addresses

192.168.0.140

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

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## 2. Nastavitev spletnega strežnika (Apache & PHP)

Na javni instanci smo namestili potrebno programsko opremo za strežbo spletnih strani in komunikacijo z bazo:

- Namestitev paketov: `sudo apt update && sudo apt install apache2 php php-mysql -y`
- Preizkus: Apache je bil uspešno zagnan, kar smo potrdili z dostopom do IP naslova v brskalniku.

## 3. Konfiguracija MariaDB baze podatkov

Na privatni instanci smo namestili MariaDB in izvedli osnovno varnostno konfiguracijo.

### Ključne nastavitve:

1. **Mrežni dostop:** V datoteki `/etc/mysql/mariadb.conf.d/50-server.cnf` smo spremenili `bind-address` na `0.0.0.0`, da baza sprejema povezave iz zunanjih IP naslovov.
2. **Podatkovna struktura:** Ustvarili smo bazo `mojapb` in tabelo `osebe`.
3. **Uporabniške pravice:** Ustvarili smo uporabnika `webuser`, ki ima pravico dostopa do baze iz kateregakoli gostitelja ('%').

```
CREATE USER 'webuser'@'%' IDENTIFIED BY 'geslo123';
GRANT ALL PRIVILEGES ON mojapb.* TO 'webuser'@'%';
FLUSH PRIVILEGES;
```

```
+-----+-----+
| User   | Host   |
+-----+-----+
| webuser | %      |
+-----+-----+
1 row in set (0.001 sec)
```

```
#
# * Basic Settings
#

#user                = mysql
pid-file             = /run/mysqld/mysqld.pid
basedir              = /usr
#datadir              = /var/lib/mysql
#tmpdir               = /tmp

# Broken reverse DNS slows down connections considerably and name resolve is
# safe to skip if there are no "host by domain name" access grants
#skip-name-resolve

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address          = 0.0.0.0
#
# * Fine Tuning
#

#key_buffer_size      = 128M
```

```

admin@ip-192-168-0-10:~$ cat /var/www/html/vnos.php
<?php
// Omogoči izpis napak za debugiranje
ini_set('display_errors', 1);
error_reporting(E_ALL);

$servername = "192.168.0.140"; // Privatni IP tvoje baze
$username = "webuser";
$password = "geslo123";
$dbname = "mojapb";

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die("Povezava ni uspela: " . $conn->connect_error);
}

// Vnos nove osebe
if ($_SERVER["REQUEST_METHOD"] == "POST" && !empty($_POST['ime']) && !empty($_POST['priimek'])) {
    $ime = $conn->real_escape_string($_POST['ime']);
    $priimek = $conn->real_escape_string($_POST['priimek']);
    $sql = "INSERT INTO osebe (ime, priimek) VALUES ('$ime', '$priimek')";
    $conn->query($sql);
}
?>

<!DOCTYPE html>
<html>
<head><title>Vaja 7 - Vnos</title></head>
<body>
    <h2>Dodaj osebo</h2>
    <form method="post">
        Ime: <input type="text" name="ime"><br>
        Priimek: <input type="text" name="priimek"><br>
        <input type="submit" value="Shrani">
    </form>

    <hr>
    <h3>Seznam oseb v bazi:</h3>
    <table border="1">
        <tr><th>ID</th><th>Ime</th><th>Priimek</th><th>Akcija</th></tr>
        <?php
        $result = $conn->query("SELECT * FROM osebe");
        while($row = $result->fetch_assoc()) {
            echo "<tr>
                <td>{$row['id']}</td>
                <td>{$row['ime']}</td>
                <td>{$row['priimek']}</td>
                <td><a href='izbrisi.php?id={$row['id']}'>Izbriši</a></td>
            </tr>";
        }
        $conn->close();
    <?>
    </table>
</body>
</html>
admin@ip-192-168-0-10:~$

```

## 4. Varnostne skupine (Security Groups)

Nastavili smo pravila požarne pregrade:

- **SG-Web-Vaja7:** Odprt port 80 (HTTP) za ves svet.
- **SG-DB-Vaja7:** Odprt port 3306 (MySQL) izključno za promet, ki prihaja iz varnostne

sg-007b4c5e74aa90f84 - SG-Web-Vaja7

Actions

Details

Security group name

SG-Web-Vaja7

Security group ID

sg-007b4c5e74aa90f84

Description

web

VPC ID

vpc-0735033c90d61e0f7

Owner

159858924832

Inbound rules count

3 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (3)

Manage tags

Edit inbound rules

Search

| <input type="checkbox"/> | Name | Security group rule ID | IP version | Type  | Protocol | Port range | Source           | Description |
|--------------------------|------|------------------------|------------|-------|----------|------------|------------------|-------------|
| <input type="checkbox"/> | -    | sgr-095fc51259ed71182  | IPv4       | SSH   | TCP      | 22         | 84.52.159.184/32 | -           |
| <input type="checkbox"/> | -    | sgr-0d250a6ec4ceba191  | IPv4       | HTTP  | TCP      | 80         | 0.0.0.0/0        | -           |
| <input type="checkbox"/> | -    | sgr-0e5c0bf79cd4d9d96  | IPv4       | HTTPS | TCP      | 443        | 0.0.0.0/0        | -           |

sg-06ec682c2e8ecbb81 - SG-DB-Vaja7

Actions

Details

Security group name

SG-DB-Vaja7

Security group ID

sg-06ec682c2e8ecbb81

Description

DB

VPC ID

vpc-0735033c90d61e0f7

Owner

159858924832

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Search

| <input type="checkbox"/> | Name | Security group rule ID | IP version | Type         | Protocol | Port range | Source                  |
|--------------------------|------|------------------------|------------|--------------|----------|------------|-------------------------|
| <input type="checkbox"/> | -    | sgr-0d18ec8bc92289cce  | -          | MySQL/Aurora | TCP      | 3306       | sg-007b4c5e74aa90f84... |
| <input type="checkbox"/> | -    | sgr-0a8ee2ceb69e581a7  | IPv4       | SSH          | TCP      | 22         | 192.168.0.10/32         |

## 5. Povezava in testiranje

Napisali smo PHP skripto vnos.php, ki se preko privatnega IP naslova poveže na bazo.

**Preizkus povezave:** Z ukazom nc -zv 192.168.0.140 3306 smo potrdili, da je port odprt in dosegljiv (succeeded).

**Končni rezultat:** V brskalniku se ob obisku naslova izpiše tabela s podatki, ki so shranjeni v privatni bazi.

Dodaj osebo

Ime:

Prezime:

Seznam oseb v bazi:

| ID | Ime  | Prezime | Akcija                  |
|----|------|---------|-------------------------|
| 1  | Anna | Hiribar | <a href="#">Izbrisi</a> |
| 3  | test | test    | <a href="#">Izbrisi</a> |